The Reciprocal-Power Hinge

From potential as $1/r$ to field strength and flux density as $1/r^2$

IB Physics A3 worksheet

Central idea. A point source produces two related radial patterns:

$$\text{potential} \propto \frac{1}{r}, \qquad \text{field strength or flux density} \propto \frac{1}{r^2}.$$

The physical constants and signs depend on the source. The power of r comes from radial geometry. Throughout, assume point sources or spherical symmetry and no absorption.

1. Potential: one reciprocal power

For a point mass and a point charge,

$$V_g = -\frac{GM}{r}, \qquad V_e = \frac{kQ}{r}.$$

Potential is energy per unit test property:

$$V_g = \frac{E_{p,g}}{m}, \qquad V_e = \frac{E_{p,e}}{q}.$$

A spacecraft moves from r to $3r$ from a planet.

Tasks

1. By what factor does the magnitude of V_g change?

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2. Does V_g increase or decrease numerically? Explain the role of the negative sign.

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3. A positive point charge produces $V_e > 0$. What happens to V_e as r increases?

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4. State the reference used in both expressions.

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Hinge question What is shared by the two expressions, and what physical information is carried by the numerator and sign?

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2. Field strength: two reciprocal powers

For the same sources,

$$g = \frac{GM}{r^2}, \qquad E = \frac{k|Q|}{r^2}.$$

A test object then experiences

$$F_g = mg, \qquad F_e = qE.$$

Tasks

1. A point is moved from r to $3r$. By what factor do g and E change?

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2. Compare this with the change in potential over the same move.

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3. Explain why electric field direction depends on the sign of Q, while gravitational field points towards the source mass.

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4. A student says, "potential and field are the same idea because both decrease with distance." Identify the missing distinction.

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Hinge question Potential describes energy per unit test property. Field strength describes force per unit test property. How does this explain the different powers of r?

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3. Spherical spreading and flux density

A source radiates power L equally in all directions. At radius r, that power is spread over a sphere:

$$A = 4\pi r^2, \qquad S = \frac{L}{4\pi r^2}.$$

Here S is irradiance: power received per unit area.

Tasks

1. By what factor does S change when distance doubles?

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2. Calculate S at $r = 2.0 \times 10^{11}$ m for $L = 3.8 \times 10^{26}$ W.

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3. Complete and interpret:

$$S(4\pi r^2) = \underline{\qquad\qquad}.$$

____

4. Explain why the irradiance decreases even though the total power crossing each complete sphere is unchanged.

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Hinge question Which quantity follows $1/r^2$: total flux, flux density, or both?

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Synthesis. Complete the scaling map:

quantity	radial dependence	change when $r \to 2r$
potential	________	________
field strength	________	________
irradiance	________	________
total power through a sphere	________	________

Complete: The $1/r^2$ dependence expresses ______________________; the numerator supplies ______________________.

The Reciprocal-Power Hinge

Teacher guide

Purpose. Students connect the radial dependence of potential, field strength and irradiance while keeping their meanings distinct. The intended architecture is

$$\text{point source or spherical symmetry} \longrightarrow \begin{cases} \text{potential} \propto 1/r, \\ \text{field or flux density} \propto 1/r^2. \end{cases}$$

No integration is required. The total power through a complete sphere remains constant in the ideal no-absorption model.

1. Potential

At $3r$,

$$|V_g(3r)| = \frac{1}{3}|V_g(r)|.$$

Because $V_g = -GM/r$, the numerical value increases towards zero as r increases: it becomes less negative.

For a positive source charge, $V_e = kQ/r$ is positive and decreases towards zero as r increases. Both expressions use zero potential at infinity.

The shared structure is $1/r$. The numerator gives source strength and coupling constant. The sign records the attractive gravitational interaction or the sign of electric source charge.

2. Field strength

At $3r$,

$$g(3r) = \frac{1}{9}g(r), \qquad E(3r) = \frac{1}{9}E(r).$$

Potential falls by a factor of 3 in magnitude; field strength falls by a factor of 9.

Electric field points away from positive charge and towards negative charge. Gravitational field points towards positive mass because ordinary gravitational sources have one sign and the interaction is attractive.

Potential is scalar energy per unit test property. Field strength is vector force per unit test property. The field indicates how rapidly potential changes with position; students may state this qualitatively without using calculus.

3. Irradiance and total flux

When distance doubles,

$$S \to \frac{1}{4}S.$$

Numerical value:

$$S = \frac{3.8 \times 10^{26}}{4\pi(2.0 \times 10^{11})^2} \approx 7.6 \times 10^2\ \mathrm{W\,m^{-2}}.$$

Also,

$$S(4\pi r^2) = L.$$

Thus the irradiance, or power per unit area, decreases as the same total power is distributed over an area growing as r^2. The total power through a complete sphere remains L.

Use terminology carefully:

- irradiance or flux density follows $1/r^2$;
- total flux or total power through a closed sphere is constant in the ideal model.

Synthesis answers

Quantity	**Dependence**	**When $r \to 2r$**
Potential	$1/r$	becomes one half in magnitude
Field strength	$1/r^2$	becomes one quarter
Irradiance	$1/r^2$	becomes one quarter
Total power through sphere	constant	unchanged

Suggested completion:

The $1/r^2$ dependence expresses spherical spreading or radial geometry; the numerator supplies the source-specific physical content.

Misconceptions to surface

- Potential and field strength are not interchangeable.
- A negative gravitational potential increases as it approaches zero.
- Flux density and total flux are different quantities.
- Inverse-square behaviour assumes a point source or spherical symmetry and no absorption for radiation.